

Ramana Gangarao

Dublin, CA | ramanagangarao04@gmail.com | (754) 275-7752 | linkedin.com/in/ramana-gangarao

Experience

AI Engineer, Innovapath INC (TalentScreen) – Dublin, CA Jan 2025 – present

- Developed AI powered talent management platform using RAG and multi-agent orchestration to standardize hiring, reduce bias, and cut time-to-hire.
- Designed a secure, multi-tenant AI architecture with strict tenant isolation across vector stores, databases, memory layers, and agent execution contexts.
- Implemented hierarchical multi-agent orchestration using Google ADK, enabling deterministic agent routing, multi-tool execution, and extensible agent lifecycles.
- Developed intent classification and routing logic to dynamically dispatch user queries to specialized agents (Recruiter, Hiring Manager, Mobility, Policy).
- Applied the planner-executor agentic pattern to decompose complex tasks into structured plans and executions, improving output accuracy and reliability.
- Designed and optimized the RAG pipeline with document chunking strategies, metadata-aware retrieval, hybrid semantic filtering, and reranking to improve retrieval precision.
- Stored end-to-end agent execution and conversational data in GCP NoSQL persistence (Firestore) to meet audit, replay, and compliance explainability requirements for hiring decisions.
- Integrated Langfuse for end-to-end tracing, capturing prompt versions, tool calls, retrieval context, agent transitions, and latency metrics.
- Conducted system-level and agent-level evaluations using task benchmarks, LLM-as-judge scoring, and ablation studies (agent removal and fault injection) to assess robustness, emergent behaviors, error propagation, and performance gains over single-agent baselines.
- Implemented hybrid memory management with short-term session memory (Redis) and long-term persistent memory using PostgreSQL and Milvus for RAG context.
- Built comprehensive guardrails including grounded-response enforcement, citation validation, bias masking, schema-validated outputs, and hallucination detection.
- Architected and deployed agentic AI services on GKE, enabling scalable, containerized orchestration of multi-agent workflows.
- Used Vertex AI for managed LLM inference, evaluation workflows, and model versioning across development and production environments.
- Implemented centralized observability and security using Cloud Logging, IAM, and Secret Manager for auditability, access control, and secrets management.

Developer Analyst I / MLOPS, Wipro Limited – Chennai, India Sept 2021 – Aug 2024

- Designed and maintained offline and online feature pipelines, including schema validation, time-based aggregations, and consistency between training and serving data.
- Implemented real-time feature updates using incremental writes and TTLs to keep features fresh and production-aligned.
- Deployed and operated real-time model serving APIs on Amazon SageMaker, managing authentication, versioned endpoints, traffic routing, and rollback strategies.
- Built scalable batch inference pipelines for large datasets with idempotent outputs and full input/output traceability.
- Integrated A/B testing and shadow deployments to safely evaluate new models against live production traffic.
- Used MLflow for experiment tracking, logging parameters and metrics, and managing model versions via the model registry.
- Automated model packaging and deployment through CI/CD pipelines, promoting models across dev, staging, and production environments.
- Implemented monitoring and alerting for model latency, error rates, and data drift to detect production issues early.
- Supported automated retraining workflows, triggering pipelines based on performance degradation or data quality signals.

- Collaborated with data science and platform teams to define model contracts, SLAs, and production readiness standards.

Software Engineer, INNOVAPATH IT Solutions PVT Ltd – Hyderabad, India

Sept 2016 – Sept 2021

- Built and deployed Spring Boot microservices using JPA/Hibernate, applying design patterns and Java 8 features for scalable backend development.
- Developed automation frameworks using Selenium, TestNG, RESTAssured, and BDD, improving UI and API test coverage and maintainability.
- Containerized Java and Python services with Docker and deployed on AWS EC2 and S3, ensuring high availability and production stability.
- Implemented CI/CD pipelines using Jenkins and Git, automating builds, tests, and deployments for faster and reliable software delivery.

Skills

Programming Languages: Python, JavaScript, Java

Frameworks: Scikit-learn, NumPy, Pandas, Pytorch, MLflow, Reactjs, Nodejs, Redux

Libraries: Google ADK, Langchain, LangGraph, Llamaindex, Langfuse , HuggingFace, Transformers

LLMs: Gemini, BERT, OpenAI

Database: SQL, PostgreSQL, MongoDB, MilvusDB, Neo4j

GCP: Vertex AI, GKE, Cloud run, Cloud Functions, Cloud Logging, Cloud Storage, Secret Manager